

A Survey of Intrusion Detection Models based on NSL-KDD Data Set

Rajesh Thomas
Abu Dhabi Men's College
Higher Colleges of Technology
Abu Dhabi
rthomas1@hct.ac.ae

Deepa Pavithran
Abu Dhabi Polytechnic
MBZ city
Abu Dhabi
deepa.pavithran@adpoly.ac.ae

Abstract—An Intrusion detection system is a key component of the security management infrastructure. Machine learning advances has benefited many domains including the security domain. Anomaly based Intrusion Detection Systems using machine learning techniques can be trained to detect even unknown attacks. In this paper we conduct a comprehensive review of various researches related to Machine Learning based IDS using the NSL-KDD data set. We propose a generic process flow for anomaly-based IDS and describe this process flow components in the context of related researches carried out. We then point out interesting future research ideas.

Keywords—Intrusion Detection, Machine Learning, NSL-KDD data set, Security

I. INTRODUCTION

Intrusion detection systems are of two types misuse-based and anomaly based. Traditional security systems for intrusion detection are misuse-based or signature-based. They rely on malware signatures to detect, trigger and log malicious activities. An example would be that of detecting a port scan. Anomaly based Intrusion Detection Systems (IDS) analyze normal behavior or situations and flag deviations as security breaches. An example of anomaly detection is that of excessive bandwidth usage or abnormal login times. To achieve optimal accuracy in anomaly-based IDS, it needs to be trained to detect abnormal behavior and differentiate it from normal behavior. This process involves a learning phase which is carried out using security data sets.

Machine Learning (ML) technologies employ mathematical algorithms to analyze big data sets and predict future values of the variables of interest. In the field of cyber security, machine learning techniques can be used to train and analyze the IDS on security related data sets. A well trained IDS can detect illegal network activity and can even predict and detect new attacks such as zero-day attacks. Anomaly based IDS can assist cyber security administrators to develop countermeasures and appropriate responses to protect assets

In this paper we review the various researches related to anomaly detection based on the Network Security Laboratory Knowledge Discovery and Data Mining (NSL-KDD) data set [1]. Further parts of this paper are organized in the following way. In section II we describe the NSL-KDD data set. Performance evaluation metrics are discussed in section III. In section IV we explain the generic flow of the research activities conducted by various researches and present a generic model of these activities. In Section V we summarize the different studies related to ML based IDS and their accuracies. In section VI, we present some ideas worth pursuing further and finally in section VII we conclude the paper.

II. NSL-KDD DATASET

The KDD99 data set which was created for the KDD Cup Challenge in 1999 was one of the most widely used data sets for cyber security research using data mining techniques. This data set is based on the DARPA 1998 data set created by the Cyber Systems and Technological group of the MIT Lincoln laboratory [2]. The data set is composed of network and operating systems data collected over 9 weeks. The first 7 weeks data represents the training set and the remaining 2 weeks of data represents the test set. KDD99 provided researchers with an opportunity to study machine learning based IDS.

Tavallae et al. [3] conducted statistical analysis on the KDDCUP'99 data set and concluded that the usage of this data set resulted in poor evaluation of anomaly detection. To solve the issues, they proposed a new data set the NSL-KDD data set, based on the original KDD data set.

A. Improvements in the NSL-KDD data set

The NSL-KDD data set has several improvements [1] to the KDD99 data set.

- Redundant records in the training set are excluded. This ensures that classification algorithms will not have to deal with the bias induced by the more frequent records.
- Duplicate records in the testing set are excluded. This improves the performance of the learning phase of the machine learning algorithms since each test record is evaluated only once. It also eliminates the classification bias induced by the frequent records.
- Total records selected from each difficulty group are inversely proportional with respect to their percentage in the original KDD data set. This results in the performance of the different machine learning methods to vary widely thereby making it more efficient in accurate evaluation of the different learning methods.
- The number of records in the training and testing sets are practical to make it affordable to run the algorithms on the complete data set without typically selecting a small sample of the data set randomly. This ensures consistency regarding the evaluation results of the work done by different researchers.

Other researchers [4],[5], [6] have shared the same views about the improvements of the NSL-KDD data set.

Although the NSL-KDD data set suffers from some problems, it is a very effective data set that can be used for research purposes [4], [5]. Moreover, it is difficult to obtain real world security data sets considering the nature of the security domain and while there are other data sets such as

UNSW-NB15 data set [6], the NSL-KDD data set is considered as one of the best ones for anomaly detection research.

B. Attacks represented in the NSL-KDD dataset

The attacks in the dataset represents four categories each of which is described in the below paragraph.

Denial of Service (DoS): In this type of attack, the threat actor sends very high number of malicious requests to a server. The machine's memory and computing resources will be too full or busy to service legitimate traffic thus denying service to the genuine users.

User to Root Attack (U2R): represents an attack type where the attacker tries to gain root or administrator privileges with the initial normal user access.

Remote to Local Attack(R2L): executed by an attacker who wants to send data to a machine over the network and fraudulently gains local access to that machine to execute the exploit.

Probing Attack: scanning a network to gain information about its details and vulnerabilities which can later be used to exploit the found vulnerabilities or execute other types of attacks.

C. Features grouping in the NSL-KDD data set

The NSL-KDD data set features are classified into 3 types

Basic features: This represents the attributes related to a TCP/IP connection.

Traffic features: this group contains features related to a single window interval and is further sub divided into same host features and same service features. These subgroups characteristically examine only the connections related to same host and same services respectively with respect to the current connection. To reflect a more real world like scenario, an examining window of 100 connections is considered in the NSL-KDD data set as compared to a time window of 2 seconds in the KDD'99 set.

Content features: attacks such as R2L and U2R does not have noticeable frequent sequential patterns. The nature of these types of attacks are embedded in the data content of the attack itself. Therefore, the data part has to be looked into for suspicious activity, e.g. the number of times the login attempt failed

D. Sample record

The data set contains 43 attributes of which 41 are from the original KDD99 data set. Additionally the 42nd attribute represents the data label and the 43rd attribute represents the level of difficulty added to the set [7]

The details of the of the NSL-KDD data set features is described by [4].

A sample record of comma separated 43 attributes from the data set is shown.

```
0,tcp,telnet,SF,121,174,0,0,0,0,1,0,0,0,0,0,0,0,0,0,0,1,1,0.0
0,0.00,0.00,0.00,1.00,0.00,0.00,255,120,0.47,0.02,0.00,0.00,
0.00,0.00,0.01,0.03,guess_passwd,12
```

III. PERFORMANCE METRICS OF IDS

The confusion matrix is used to depict the actual and predicted classes in cybersecurity attacks. The confusion matrix is represented by the following terms.

True Positive (TP): the instance is correctly predicted as an attack.

True Negative (TN): correctly predicted as a non-attack or normal instance.

False Positive (FP): a normal instance is wrongly predicted as attacks.

False Negative (FN): an actual attack is wrongly predicted as non-attacks or normal instance.

False positives where a normal network activity is classified as an attack can waste the valuable time of security administrators. False negatives have the worst impact on organizations since an attack is not detected at all.

TABLE I. CONFUSION MATRIX FOR SECURITY ATTACK CLASSIFICATION

		Predicted class	
		<i>Attack</i>	<i>Normal</i>
Actual class	<i>Attack</i>	TP	FN
	<i>Normal</i>	FP	TN

Using the confusion matrix shown in table 1, the commonly used metrics for evaluation are, DR, FAR and Accuracy.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

$$\text{Attack detection rate} = \frac{TP}{TP + FN}$$

$$\text{False Alarm rate} = \frac{FP}{FP + TN}$$

The diagonal of the matrix represents the correct predictions whereas the other elements show the incorrect predictions[8].

The receiver operating characteristics (ROC) is another metric that is used for the analysis of the results. It can be used to determine the cost sensitivity of a classifier [5],[9].

IV. GENERIC THEME OF RESEARCHES

We group the various researches surveyed into themes based on the training and testing data used, the preprocessing implemented, the ML algorithms used and the results of the respective models. By analyzing these themes, we present a generic flow of the research activity as shown in Fig. 1. The data set is split into training and test sets and pre-processing is carried out on the data. Pre-processing comprises of normalization and feature selection. Additionally, conversion

of attributes to other suitable formats such as strings to numeric can be employed in the pre-processing stage. Machine learning algorithms are applied to the pre-processed data in the core model part. The model core is developed and trained on the training data set. After the training is completed, the model is evaluated by testing it on the test data to give the final classified results. The types of algorithms employed in the model core will vary and contribute to different anomaly detection rates. Finally, the performance of the model core is then evaluated with other models.

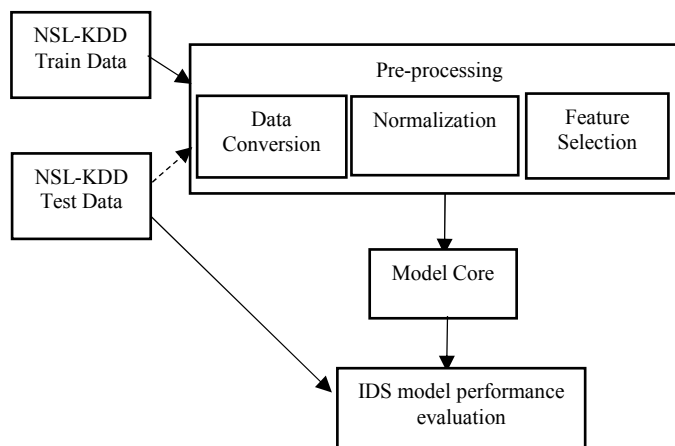


Fig. 1. Generic flow of research activities for intrusion detection using NSL-KDD data set.

A. Training and Testing Data

The NSL-KDD data set provides two sets of training sets, the complete training set which includes the attack type labels and difficulty and a 20% subset of the complete training set. The testing set is also available likewise.

Dhanabal and Shantharajah [4] used 20% of the NSL-KDD data set in their experiments that was conducted with the automated data mining tool, WEKA

In the proposed model by [5] 80% instances of the data set was used for training while the remaining 20% was used for testing purpose. The 20% test data contains 25192 instances of which 13449 are benign data and 11743 are attack data.

Ingre and Yadav [10], selected 18718 records for the training part, out of which 17672 were chosen at random. Training was conducted on the full features dataset as well as the reduced feature dataset. It was observed that the training and testing took more time for the full feature 41 attribute as compared to the 29-attribute set.

To summarize this part, we found that most researchers have used the 20% training data set.

B. Pre-processing techniques

Before machine learning algorithms can be applied to the data, it needs to be converted into a format that is suitable for data analysis by the chosen ML algorithm. Pre-processing techniques employed directly contribute towards the efficiency of the overall system. Pre-processing is a typically combination of data conversion, normalization and feature selection techniques. This part of the research activity is a

very important step towards attaining increased detection rates by the model core subsystem.

1) Feature selection

The high dimensionality of the data set creates challenges in analyzing the data, therefore feature selection or dimension reduction techniques are used for data analysis. Using feature selection methods, a subset of the features that aid in optimal performance of the system are selected by applying certain evaluation criteria. This reduces the computation time and model complexity of the system and improves the accuracy of the classification.

Correlation based feature selection method was used by [4] to reduce the dimensionality of the features from 41 to 6.

Pre-processing of training and testing data file was done by [11] to generate 14 new training data files based on a combination of the 4 classes such as BCTH, BCT, BCH, etc.

Deshmukh, Ghorpade and Padiya [8] employed Fast Correlation Based Filer (FCBF) algorithm to reduce the dimensionality of the data set in the pre-processing stage. Final pre-processing task of discretization is performed by Equal width discretization technique.

Rai, Devi and Guleria [12] employed feature selection using information gain technique in their research. The information gain of all the features was computed and only the 16 attributes whose information gain was greater than the average information gain was selected.

Ingre and Yadav [10] converted non numeric data for attributes such as protocol type, service and flag to numerical format to make it compatible for providing as an input to the ANN (Artificial Neural Network). Class attributes such as normal, DoS, probe, R2L, and U2R were given the values of 1,2,3,4 and 5 respectively. These class attributes were then converted to bit form as 10000, 01000, 00100, 00010 and 00001 respectively. The position of 1 in the bit representation indicates the targeted class.

Aljawarneh, Aldwairi and Yassein [5], converted the non-numerical values of the features 2,3 and 4 representing the protocol type, service and flag respectively to numerical values. The numerical values assigned were TCP=1, UDP=2, ICMP=3. Similarly, the different attack types such as DoS, Probe, R2L and U2R are represented in numerical format. Using information gain (IG), features with IG greater than 0.4 was used thereby reducing the features set from 41 to 8.

Parsaei, Rostami and Javidan [13], reduced the 41-feature dataset to 21 features using the Leave One Out (LOO) method. This technique evaluates the importance of each feature based on accuracy and false positive rate. The training set was sampled 10 times by changing random generator seed and each of these times the synthetic minority oversampling technique (SMOTE) method was used to balance the data set and cluster center nearest neighbor (CANN) was used to classify the dataset and build the model.

2) Normalization

Since certain classifiers yield better accuracy on normalized data, the data set was pre-processed and normalized in the range of 0 to 1 [4].

Normalization was applied by [10] to the attribute values using **z-score normalization technique**. The mean and standard deviation after normalization is equal to 0 and 1 respectively.

Min max normalization was used by [5] to normalize the data so that values of the features can be represented in the 0-1 range. Feature selection was achieved by search method and sub set attribute vector.

C. Core model

This part of the system represents the techniques that form the core of system. Different classification techniques can be employed in this subsystem.

Aggarwal and Sharma [11] grouped the 41 attributes into four classes namely – Basic (B), Content (C), Traffic (T) and Host(H). The NSL-KDD data set was analyzed from the viewpoint of these four classes. Random tree binary classifier using the WEKA tool was used. Random tree classifier is an ensemble of forest containing tree predictors. Every tree in the forest is applied in the classification process and the output of the label will be that of the class label with the majority of the votes[11]. The results of the experiment show that the presence of the basic class attributes has the maximum Detection Rate (DR) and the traffic attributes show lower DR. It was also observed that False Alarm Rate (FAR) was comparatively higher when content class attributes were included, and the host class attributes showed the best FAR

Dhanabal and Shantharajah [4] studied the relationship of the network protocols associated with the attack type. The dataset was categorized based on the previously mentioned four types of attacks. J48, SVM and Naïve Bayes algorithms was used for classification. It was observed that when CFS was used for dimensionality reduction, J48 classifier has a better accuracy rate. Application of CFS reduces the detection time and increases the accuracy rate. In relations to the protocols, it was observed that the majority of the attacks exploited the vulnerabilities of the TCP protocol.

Shrivastava, Sondhi and Ahirwar [14] proposed a conceptual model for intrusion detection based on the machine learning techniques. Classification was used for categorizing intrusions from normal traffic. The model was tested on the basis of Accuracy, Error rate, Detection rate and False Alarm rate.

Duque and Omar [7] proposed the k-means unsupervised machine learning technique as the core of the IDS. K-Means clustering is a type of machine learning technique based on the centroid technique that partitions the data set into k partitions. This technique is used to identify outliers which represents anomaly behavior in cyber-attacks. The study was conducted using different cluster sizes. It was observed that the best results were yielded when the number of clusters was equal to the number of the data types in the data set. It was also observed that for the 22 sized cluster, the false alarms represented by False Positive Rate (FPR) is significantly lower at 4.03% than the False Negative Rate (FNR) for all tested cluster sizes.

Deshmukh, Ghorpade and Padiya [8] used classifiers such as Naïve Bayes, HiddenNaive Bayes and NBTree for the model core. The results shows that NBTree algorithm

performs well in Accuracy and Error rate as compared to the other algorithms used in the study.

Rai, Devi and Guleria [12] proposed the C4.5 decision tree approach to serve as the model core. It addresses the two key issues of feature selection and split value. The split value is taken as the average of the values in the domain of an attribute at each node. The advantage of this method is that it reduces the most frequent attribute bias since uniform weightage is given to all values in the domain. The analysis of the results revealed that the TPR of the proposed algorithm is better than C4.5 technique, although the CART algorithm had the best TPR. CART, however takes longer time in building the model. The efficiency depends on the data set size and the number of features selected for construction the decision tree. By improving the split selection, the detection efficiency of IDS can be increased.

Ingre and Yadav [10] analyzed the performance of NSL-KDD data set using Artificial Neural Networks. ANN consists of interconnected neurons that learn through a training phase and use the learned techniques to detect intrusion in unlabeled data set. Training and testing is carried out on the set using a reduced 29 feature set or the complete 41 features. The parameters such as the number of neurons, number of layers in case of multilayer ANN, the algorithm and the transfer function for the neural network needs to be selected. The transig transfer function, Levenberg-Marquardt and BFGS quasi-Newton backpropagation algorithm for updating the weight and bias was used in their research. The accuracy of the Levenberg-Marquardt algorithm with 21 hidden layers was found to be at 99.3% and for the other afore mentioned algorithms it was 98.9%. It was observed that the binary class classification gives higher accuracy of attack detection.

Aljawarneh, Aldwairi and Yassein [5] formulated a **hybrid model** for anomaly-based IDS. The pre-processed and normalized data set is analyzed by using various classifiers such as J48, Meta Pagging, Random Tree, REPTree, AdaBoostM1, Decision Stump and Naïve Bayes. Using **VOTE scheme and Information gain** the classifier that yields the best accuracy was chosen for feature selection. The results indicate the **highest classification percentage of 99.81 for the proposed hybrid model**. Additionally, it also has the **lowest false positive rate and highest true positive rate (TPR)**. Analysis of the results also point towards the fact that the majority of the attacks are done using the TCP protocol's weaknesses.

Parsaei, Rostami and Javidan [13] employed a hybrid approach proposed by [15] that combines SMOTE and CANN to improve the detection rate of low frequency attacks like R2L and U2R. The number of U2R and R2L class instance was increased using SMOTE. This balances the number of instances of each type of attack in the dataset. CANN method a twostep process. In the first step cluster centroids is computed by using k-means. In the second step the distances between each data point with respect to the cluster centroids and with respect to their nearest neighbor is summed. The results indicated a greater performance as compared to the baseline method in terms of intrusion detection rate. However, the accuracy and false alarm rate achieved was lower.

V. SUMMARY OF ML TECHNIQUES USED IN IDS

The detection rates vary depending on the preprocessing technique and the core ML algorithm used by the different researchers. Table II summarizes the various studies reviewed by us and their overall efficiencies achieved by the different researchers.

TABLE II. SUMMARY OF THE REVIEW

Reference	Year	Research Paper Title	Algorithm used in pre-processing / model core	Accuracy %
[4] D. H. Deshmukh, T. Ghorpade, and P. Padiya	2015	Improving classification using preprocessing and machine learning algorithms on NSL-KDD dataset	Naïve Bayes HiddenNaive Bayes NBTree	88.20 93.40 94.60
[5] S. Aljawarneh, M. Aldwairi, and M. B. Yassein	2016	Anomaly-based intrusion detection system through feature selection analysis and building hybrid efficient model	Hybrid comprising of J48 Random Tree Naïve Bayes	99.81
[6] P. Aggarwal and S. K. Sharma	2015	Analysis of KDD Dataset Attributes - Class wise for Intrusion Detection	Ensemble Random Tree	Varies from 70.29 to 86.21 for different class combinations
[7] S. Duque and M. N. Bin Omar	2015	Using Data Mining Algorithms for Developing a Model for Intrusion Detection System (IDS)	k-means	81.61 for cluster size 22
[8] L. Dhanabal and S. P. Shantharajah	2015	A Study on NSL-KDD Dataset for Intrusion Detection System Based on Classification Algorithms	CFS J48 SVM Naïve Bayes	Varies from 70.1 to 99.8 for different attack types and algorithms
[10] K. Rai, M. S. Devi, and A. Guleria	2016	Decision Tree Based Algorithm for Intrusion Detection	C4.5 Decision Tree Split (DTS)	79.52
[11] B. Ingre and A. Yadav	2015	Performance analysis of NSL-KDD dataset using ANN	Levenberg-Marquardt (LM) algorithm and with 21 hidden	99.3 98.9

			layers BFGS quasi-Newton Backpropagation algorithm and 23 hidden layers	
[12] M. R. Parsaei, S. M. Rostami, and R. Javidan	2016	A Hybrid Data Mining Approach for Intrusion Detection on Imbalanced NSL-KDD Dataset	SMOTE CANN	98.99

VI. FUTURE RESEARCH IDEAS

The future research areas are related to the pre-processing and model core part of the research flow activities represented in Fig. 1. We believe the following research ideas are worth pursuing further.

- Using k-means as a level 1 detection and signature based detection for level 2 detection [7].
- Improving split in k-means for classification purposes using techniques such as geometry mean [12]
- Further research in the application of optimization techniques and implementation of distributed network IDS [5].
- Exploring new pre-processing techniques that can improve the model core.
- Since the number of clusters significantly impacts the efficiency [7], knowing the number of clusters before the application of the k-means can prove beneficial. K-means followed by signature-based approach could help reduce the FNR. Techniques for automatically identifying the number of clusters would be a good future research direction.
- Use dual data sets such as the newly published UNSW-NB15 data set and the NSL-KDD data set described in this paper for intrusion detection analysis. Comparisons and conclusions can be drawn on the effectiveness of the ML algorithms on these two different data sets.
- Design of a hybrid intrusion detection framework based on ML anomaly detection with seamless integration into the standard signature-based IDS.

VII. CONCLUSION

In this paper we have surveyed the various researches conducted for anomaly-based intrusion detection using the NSL-KDD data set. We have outlined the flow of the research activities and represented them in a generic process model. New researchers should focus on the pre-processing and model core parts shown in our generic process flow, since the key areas of research in improving the detection rate in IDS are in these stages. Pre-processing is very important as it directly affects the accuracy of the classifiers.

Pre-processing coupled with ANN shows some of the most successful detection rates. Various studies have indicated an improvement in detection rates with the different pre-processing techniques combined with ML classification techniques. Hybrid models too look promising for further research.

REFERENCES

- [1] "NSL-KDD | Datasets | Research | Canadian Institute for Cybersecurity | UNB," 2017. [Online]. Available: <http://www.unb.ca/cic/datasets/nsf.html>. [Accessed: 04-May-2018].
- [2] A. Buczak and E. Guven, "A survey of data mining and machine learning methods for cyber security intrusion detection," *IEEE Commun. Surv. Tutorials*, vol. PP, no. 99, p. 1, 2015.
- [3] M. Tavallaei, E. Bagheri, W. Lu, and A. A. Ghorbani, "A detailed analysis of the KDD CUP 99 data set," in *IEEE Symposium on Computational Intelligence for Security and Defense Applications, CISDA 2009*, 2009, no. Cisd, pp. 1–6.
- [4] L. Dhanabal and S. P. Shantharajah, "A Study on NSL-KDD Dataset for Intrusion Detection System Based on Classification Algorithms," *Int. J. Adv. Res. Comput. Commun. Eng.*, vol. 4, no. 6, pp. 446–452, 2015.
- [5] S. Aljawarneh, M. Aldwairi, and M. B. Yassein, "Anomaly-based intrusion detection system through feature selection analysis and building hybrid efficient model," *J. Comput. Sci.*, vol. 25, pp. 152–160, 2016.
- [6] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems (UNSW-NB15 network data set)," in *2015 Military Communications and Information Systems Conference, MilCIS 2015 - Proceedings*, 2015.
- [7] S. Duque and M. N. Bin Omar, "Using Data Mining Algorithms for Developing a Model for Intrusion Detection System (IDS)," in *Procedia Computer Science*, 2015, vol. 61, pp. 46–51.
- [8] D. H. Deshmukh, T. Ghorpade, and P. Padiya, "Improving classification using preprocessing and machine learning algorithms on NSL-KDD dataset," in *Proceedings - 2015 IEEE International Conference on Communication, Information and Computing Technology, ICCICT 2015*, 2015.
- [9] G. Kumar, "Evaluation Metrics for Intrusion Detection Systems -A Study," *Int. J. Comput. Sci. Mob. Appl.*, vol. 2, no. 11, pp. 11–17, 2014.
- [10] B. Ingre and A. Yadav, "Performance analysis of NSL-KDD dataset using ANN," *Int. Conf. Signal Process. Commun. Eng. Syst. - Proc. SPACES 2015*, Assoc. with IEEE, pp. 92–96, 2015.
- [11] P. Aggarwal and S. K. Sharma, "Analysis of KDD Dataset Attributes - Class wise for Intrusion Detection," in *Procedia Computer Science*, 2015, vol. 57, pp. 842–851.
- [12] K. Rai, M. S. Devi, and A. Guleria, "Decision Tree Based Algorithm for Intrusion Detection," vol. 2834, pp. 2828–2834, 2016.
- [13] M. R. Parsaei, S. M. Rostami, and R. Javidan, "A Hybrid Data Mining Approach for Intrusion Detection on Imbalanced NSL-KDD Dataset," *Int. J. Adv. Comput. Sci. Appl.*, vol. 7, no. 6, pp. 20–25, 2016.
- [14] A. Shrivastava, J. Sondhi, and S. Ahirwar, "Cyber attack detection and classification based on machine learning technique using nsf kdd dataset," *Int. Reserach J. Eng. Appl. Sci.*, vol. 5, no. 2, 2017.
- [15] W. C. Lin, S. W. Ke, and C. F. Tsai, "CANN: An intrusion detection system based on combining cluster centers and nearest neighbors," *Knowledge-Based Syst.*, vol. 78, no. 1, pp. 13–21, 2015.